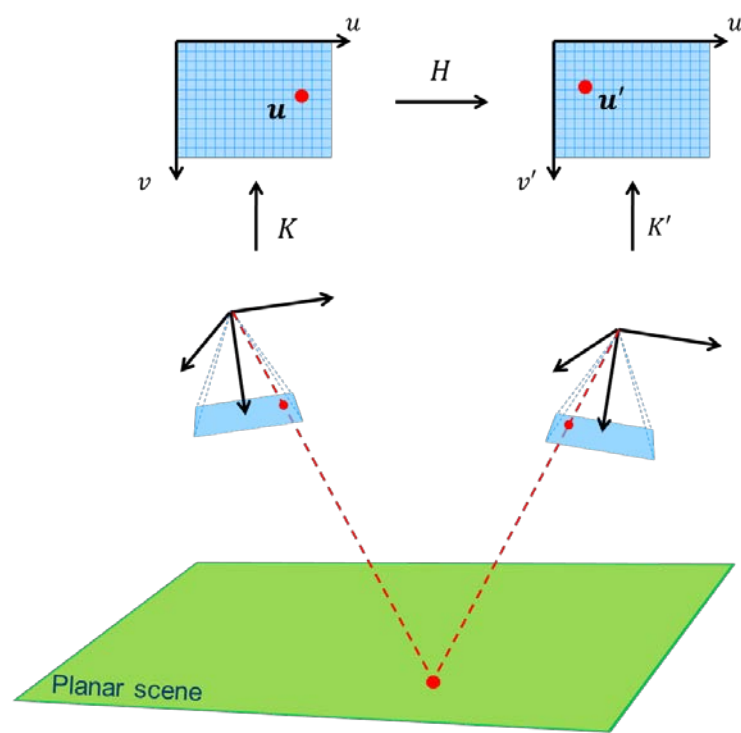


Lecture 3.3

Robust estimation with RANSAC

Motivation



- If two perspective cameras capture an image of a planar scene, their images are related by a homography H

Motivation



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Motivation



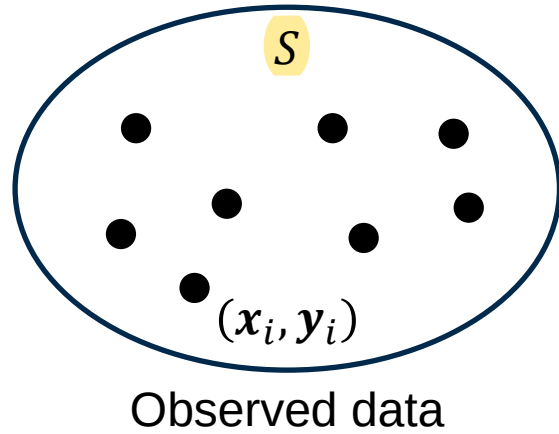
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Motivation



- If two perspective cameras capture an image of a planar scene, their images are related by a homography H
- It can be estimated if we know at least 4 point-correspondences $\mathbf{u}_i \leftrightarrow \mathbf{u}'_i$
- Correspondences can be found automatically, but typically some of them will be wrong
- A robust estimation method provides a good estimate of H despite the presence of these wrong correspondences

RANdom SAmple Consensus - RANSAC



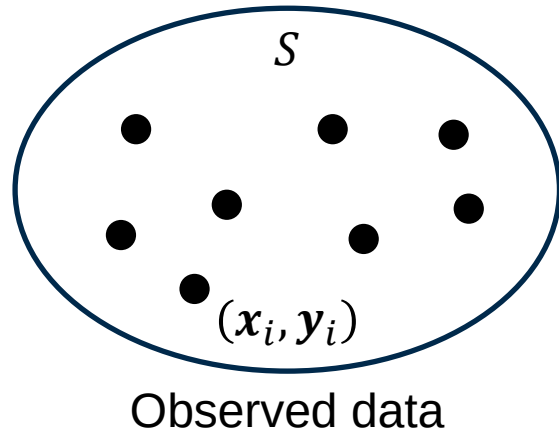
→
Estimation
method

$$y = f(x; \alpha)$$

Mathematical model
with parameters
 $\alpha = (\alpha_1, \dots, \alpha_n)$

- RANSAC is an iterative method for estimating the parameters of a mathematical model from a set of observed data containing outliers

RANdom SAmple Consensus - RANSAC



→
Estimation
method

$y = f(x; \alpha)$
Mathematical model
with parameters
 $\alpha = (\alpha_1, \dots, \alpha_n)$

$$y = ax + b$$

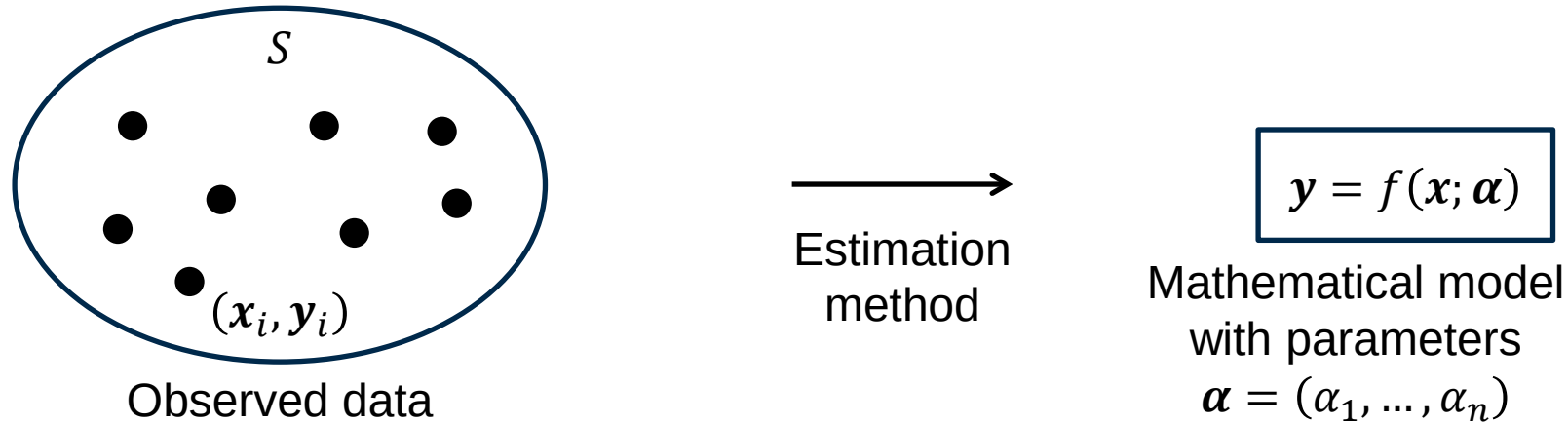
$$ax + by + c = 0$$

$$y = ax^2 + bx + c$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

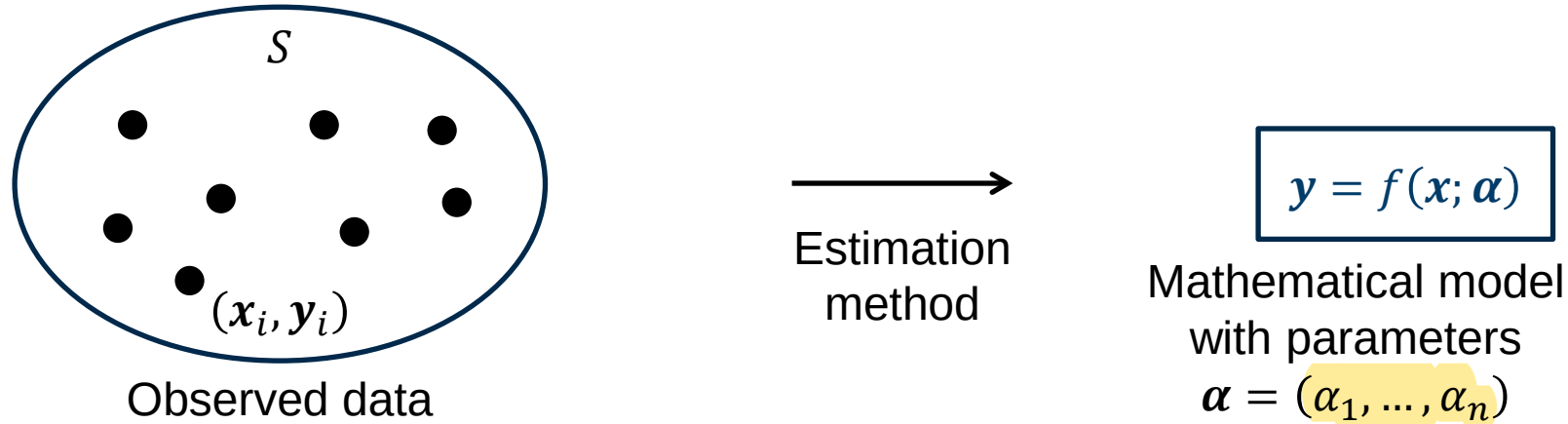
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RANdom SAmple Consensus - RANSAC



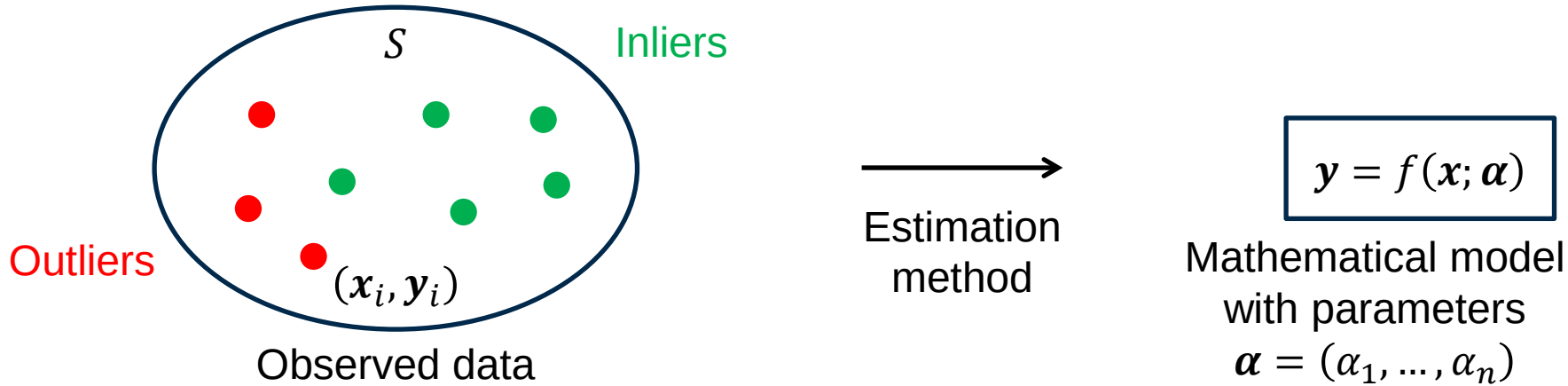
- RANSAC is an iterative method for estimating the parameters of a mathematical model from a set of observed data containing outliers
 - Robust method (handles up to 50% outliers)

RANdom SAmple Consensus - RANSAC



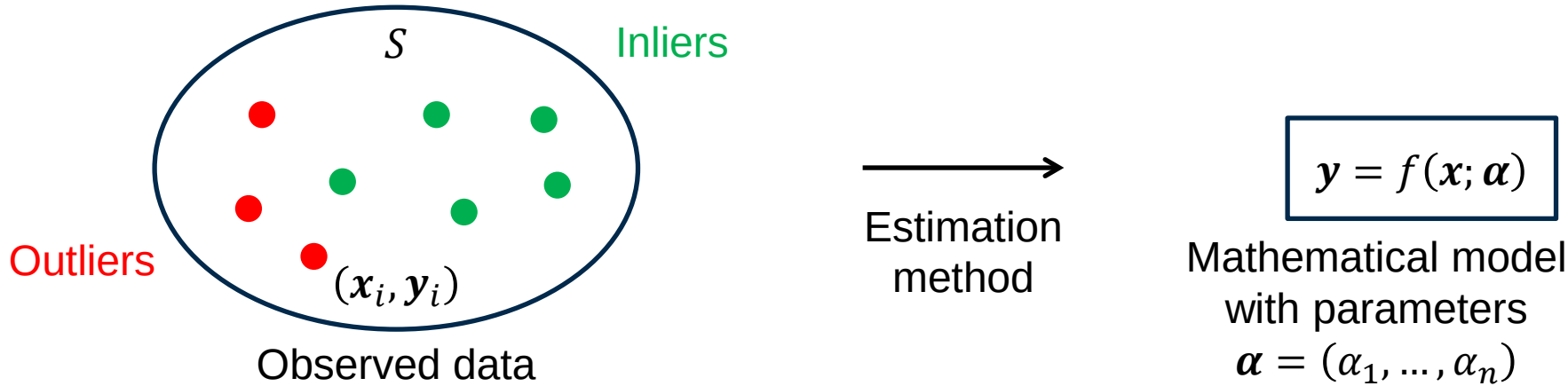
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RANdom SAmple Consensus - RANSAC



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RANdom SAmple Consensus - RANSAC



- RANSAC is an iterative method for estimating the parameters of a mathematical model from a set of observed data containing outliers
 - Robust method (handles up to 50% outliers)
 - The estimated model is random but reasonable
 - The estimation process divides the observed data into inliers and outliers
 - Usually an improved estimate of the model is determined based on the inliers using a less robust estimation method, e.g. least squares

Basic RANSAC

Objective

Robustly fit a model $y = f(x; \alpha)$ to a data set $S = \{x_i\}$

Algorithm

1. Determine a test model $y = f(x; \alpha_{tst})$ from n random data points $\{x_1, x_2, \dots, x_n\}$
2. Check how well each individual data point in S fits with the test model
 - Data points within a distance t of the model constitute a set of inliers $S_{tst} \subseteq S$
 - Data points outside a distance t of the model are outliers
3. If S_{tst} is the largest set of inliers encountered so far, we keep this model
 - Set $S_{IN} = S_{tst}$ and $\alpha = \alpha_{tst}$
4. Repeat steps 1-3 until N models have been tested

Basic RANSAC

Comments

- The number of random samples, n , is typically the smallest number of data points required to estimate the model
- Assuming Gaussian noise in the data, the threshold value t should be in the region of 2σ where σ is the expected noise in the data set
- The maximal number of tests, N , can be chosen according to how certain we want to be of sampling at least one data set $\{x_1, x_2, \dots, x_n\}$ with no outliers
- If p is the desired probability of sampling at least one n -tuple with no outliers and ω is the probability of a random data point to be an inlier, then

$$N = \frac{\log(1 - p)}{\log(1 - \omega^n)}$$

Basic RANSAC

Comments

- Standard value $p = 0.99$
- We rarely know the ratio of inliers in our set of data points, so in most situations, ω is unknown
- Instead of maximizing ω , leading to a larger than necessary N , we can modify RANAC to adaptively estimate N as we perform the iterations

		ω				
n	N	90	80	70	60	50
	2	3	5	7	11	17
	3	4	7	11	19	35
	4	5	9	17	34	72
	5	6	12	26	57	146
	6	7	16	37	97	293
	7	8	20	54	163	588
	8	9	26	78	272	1177

$$N = \frac{\log(1 - p)}{\log(1 - \omega^n)}$$

$$p = 0.99$$

Adaptive RANSAC

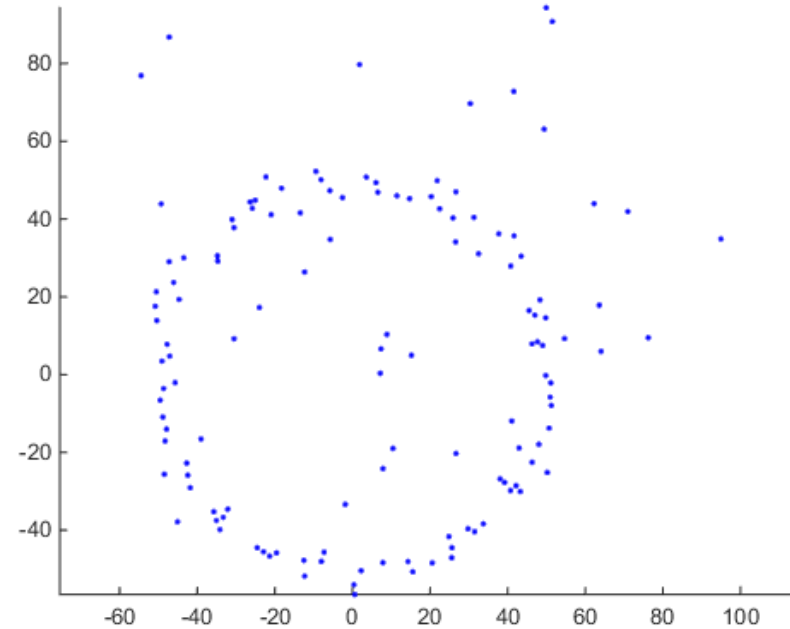
Objective

Robustly fit a model $\mathbf{y} = f(\mathbf{x}; \boldsymbol{\alpha})$ to a data set $S = \{\mathbf{x}_i\}$

Algorithm

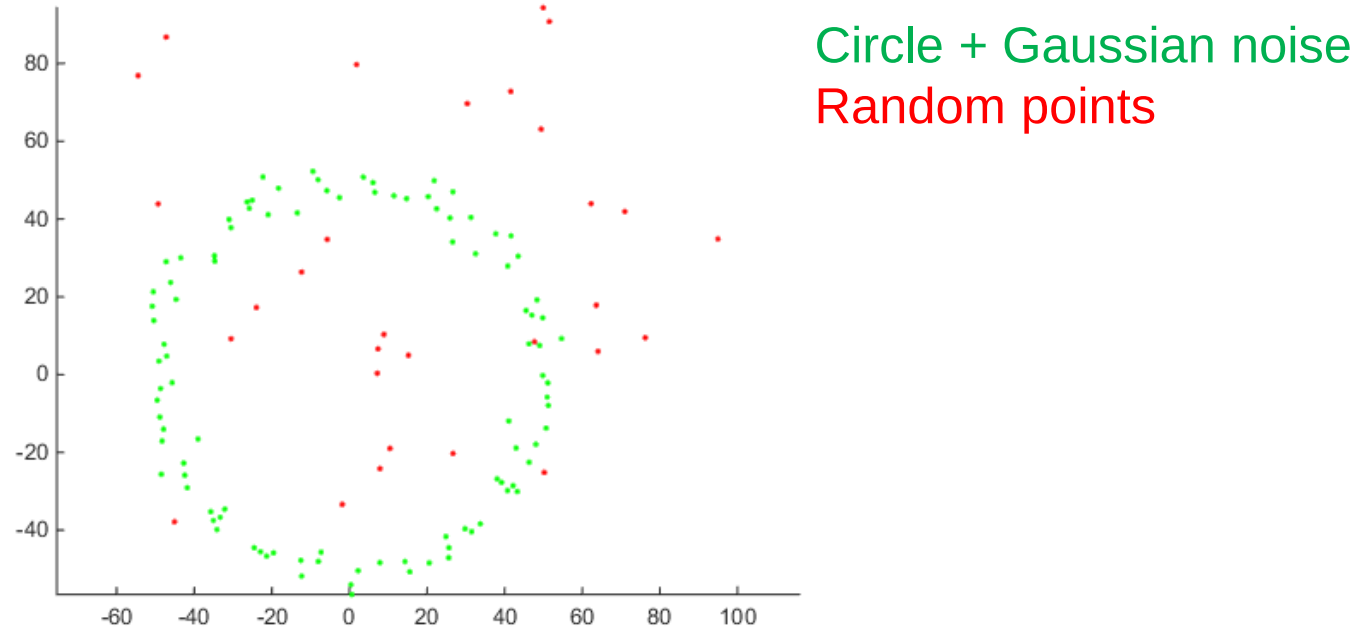
1. Let $N = \infty$, $S_{IN} = \emptyset$
2. As long as the number of iterations are smaller than N repeat steps 3-5
3. Determine a test model $\mathbf{y} = f(\mathbf{x}; \boldsymbol{\alpha}_{tst})$ from n random data points $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$
4. Check how well each individual data point in S fits with the test model
 - Data points within a distance t of the model constitute a set of inliers $S_{tst} \subseteq S$
5. If S_{tst} is the largest set of inliers encountered so far, we keep this model
 - Set $S_{IN} = S_{tst}$ and $\boldsymbol{\alpha} = \boldsymbol{\alpha}_{tst}$
 - Compute $N = \frac{\log(1-p)}{\log(1-\omega^n)}$ using that $\omega = \frac{|S_{IN}|}{|S|}$ and $p = 0.99$

Example



- Fit a circle $(x - x_0)^2 + (y - y_0)^2 = r^2$ to these data points by estimating the 3 parameters x_0 , y_0 and r

Example



- Fit a circle $(x - x_0)^2 + (y - y_0)^2 = r^2$ to these data points by estimating the 3 parameters x_0 , y_0 and r
- The data consists of some points on a circle with Gaussian noise and some random points

Example

Least-squares approach

Separate observables from parameters:

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

$$x^2 - 2xx_0 + x_0^2 + y^2 - 2yy_0 + y_0^2 = r^2$$

$$2xx_0 + 2yy_0 + r^2 - x_0^2 - y_0^2 = x^2 + y^2$$

$$\begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} 2x_0 \\ 2y_0 \\ r^2 - x_0^2 - y_0^2 \end{bmatrix} = \begin{bmatrix} x^2 + y^2 \end{bmatrix}$$

$$\begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} x^2 + y^2 \end{bmatrix}$$

So for each observation (x_i, y_i) we get one equation

$$\begin{bmatrix} x_i & y_i & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} x_i^2 + y_i^2 \end{bmatrix}$$

From all our N observations we get a system of linear equations

$$\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots \\ x_N & y_N & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \vdots \\ x_N^2 + y_N^2 \end{bmatrix}$$

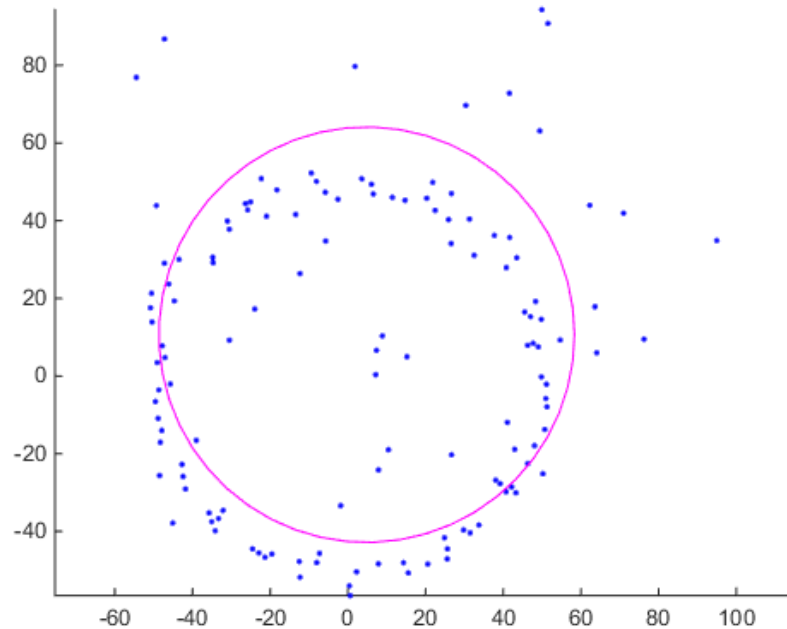
$$A\mathbf{p} = \mathbf{b}$$

Example

- One way of solving the equation $A\mathbf{p} = \mathbf{b}$ is to take the pseudo inverse $\mathbf{p} = (A^T A)^{-1} A^T \mathbf{b}$
 - This give us the solution that minimizes $\|A\mathbf{p} - \mathbf{b}\|$

Example

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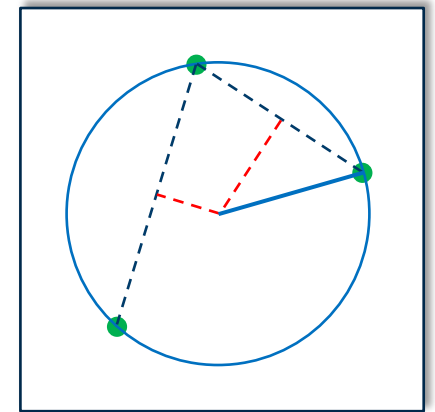
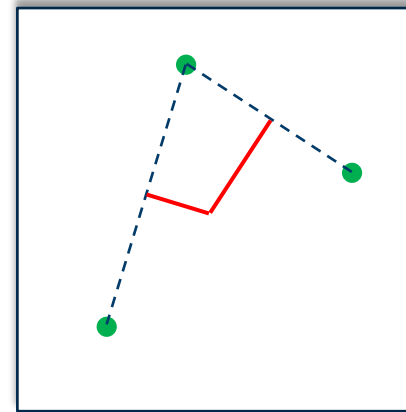
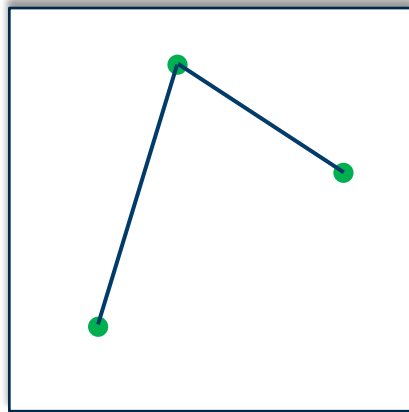
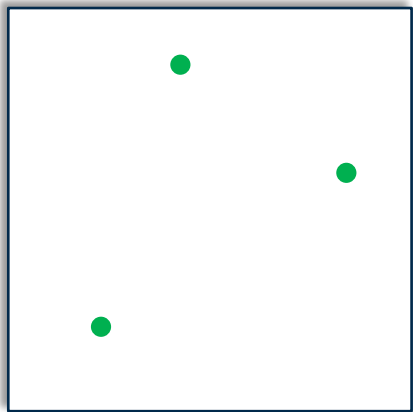
- NOT GOOD! All points are treated equally, so the random points shifts the estimated circle away from the desired solution

Example

- To estimate the circle using RANSAC, we need two things
 1. A way to estimate a circle from n points, where n is as small as possible
 2. A way to determine which of the points are inliers for an estimated circle

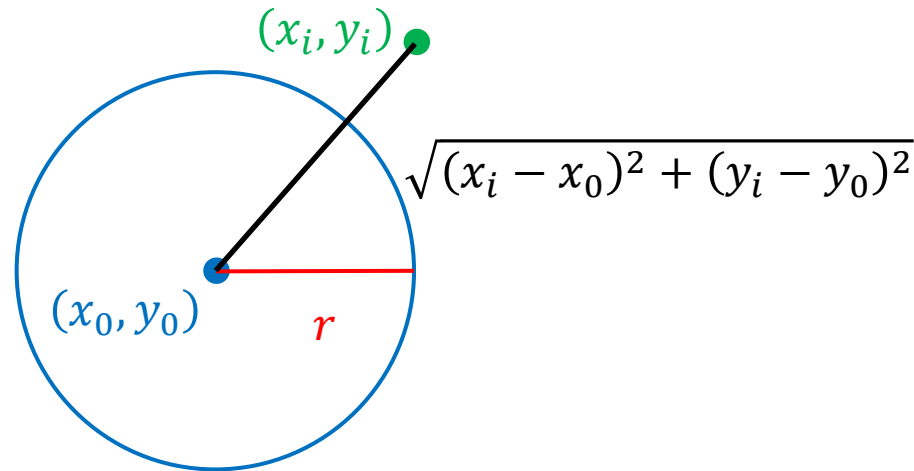
Example

- To estimate the circle using RANSAC, we need two things
 1. A way to estimate a circle from n points, where n is as small as possible
 2. A way to determine which of the points are inliers for an estimated circle
- The smallest number of points required to determine a circle is 3, i.e. $n = 3$, and the algorithm for computing the circle is quite simple



Example

- To estimate the circle using RANSAC, we need two things
 1. A way to estimate a circle from n points, where n is as small as possible
 2. **A way to determine which of the points are inliers for an estimated circle**
- The distance from a point (x_i, y_i) to a circle $(x - x_0)^2 + (y - y_0)^2 = r^2$ is given by
$$\left| \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2} - r \right|$$



Example

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$$\left| \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2} - r \right|$$
- So for a threshold value t , we say that (x_i, y_i) is an inlier if $\left| \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2} - r \right| < t$

Example

Objective

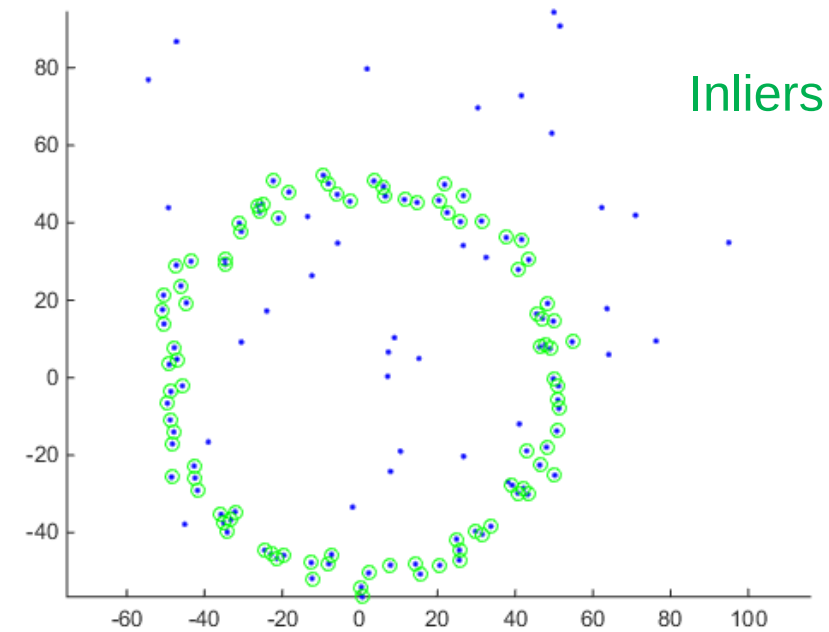
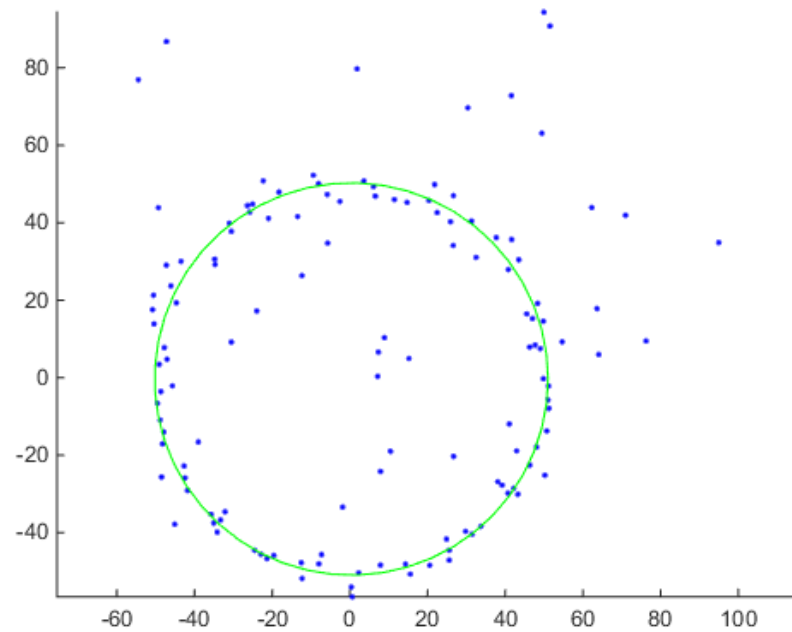
To robustly fit the model $(x - x_0)^2 + (y - y_0)^2 = r^2$ to our data set $S = \{(x_i, y_i)\}$

Algorithm

1. Let $N = \infty$, $S_{IN} = \emptyset$, $p = 0.99$, $t = 2 \cdot \text{expected noise}$
2. As long as the number of iterations are smaller than N repeat steps 3-5
3. Determine parameters $(x_{tst}, y_{tst}, r_{tst})$ from three random points from S
4. Check how well each individual data point in S fits with the test model
$$S_{tst} = \left\{ (x_i, y_i) \in S \text{ such that } \left| \sqrt{(x_i - x_{tst})^2 + (y_i - y_{tst})^2} - r_{tst} \right| < t \right\}$$
5. If S_{tst} is the largest set of inliers encountered so far, we keep this model
 - Set $S_{IN} = S_{tst}$ and $(x_0, y_0, r) = (x_{tst}, y_{tst}, r_{tst})$
 - Recompute $N = \frac{\log(1-p)}{\log(1-\omega^n)}$ using that $\omega = \frac{|S_{IN}|}{|S|}$

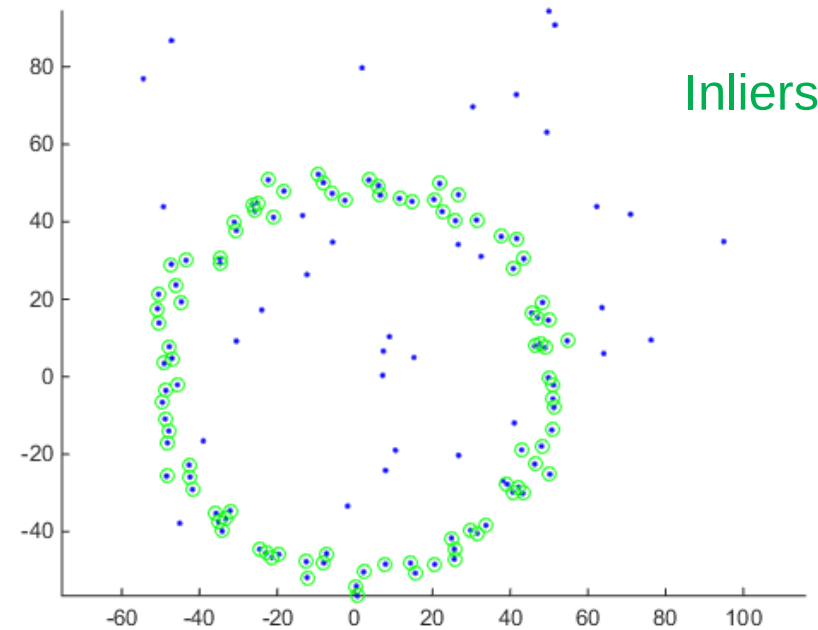
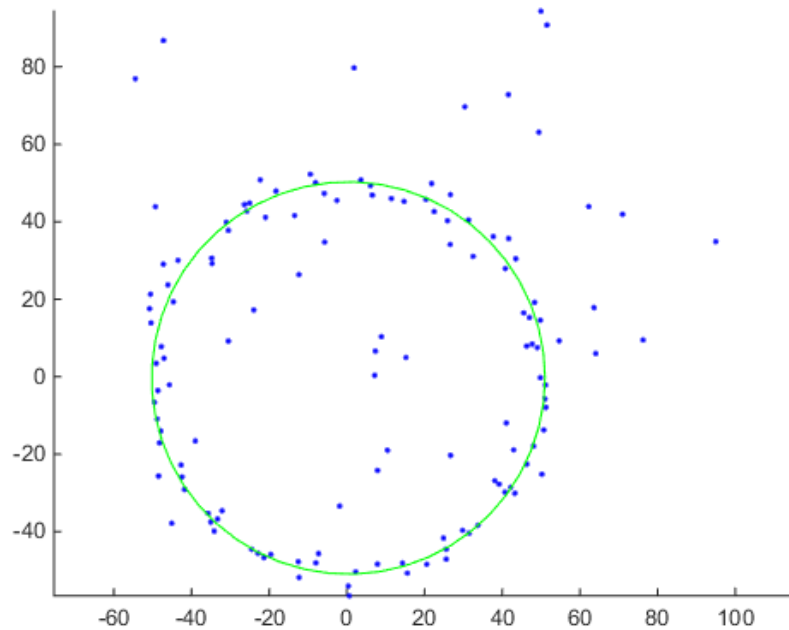
Example

- The RANSAC algorithm evaluates many different circles and returns the circle with the largest inlier set



Example

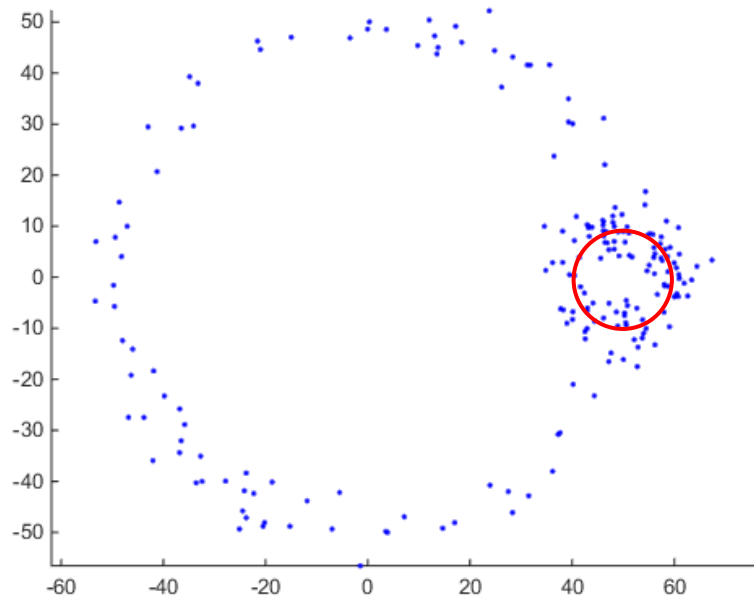
- The RANSAC algorithm evaluates many different circles and returns the circle with the largest inlier set



- An improved estimate for the circle can be found from the set of inliers using a less robust algorithm e.g. least squares

Robust estimation

- But RANSAC is not perfect...



- Several other robust estimation methods exist
 - Least Median Squares (LMS)
 - Preemptive RANSAC
 - PROgressive Sample and Consensus (PROSAC)
 - M-estimator Sample and Consensus (MSAC)
 - Maximum Likelihood Estimation Sample and Consensus (MLESAC)
 - Randomized RANSAC (R-RANSAC)
 - KALMANSAC
 - +++

Summary

- RANSAC
 - A robust iterative method for estimating the parameters of a mathematical model from a set of observed data containing outliers
 - Separates the observed data into “inliers” and “outliers” which is very useful if we want to use better, but less robust, estimation methods
- Additional reading
 - Szeliski: 6.1.4
- Homework?
 - Implement a RANSAC algorithm for estimating a line